Jack Norris

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EDUCATION

College of Computing, Georgia Institute of Technology — **Atlanta, GA**
***Degree**:* M.S. Computational Science & Engineering — *Expected Graduation 2028*

College of Engineering, University of Arkansas (UARK) — **Fayetteville, AR**
***Degree**:* B.S. Computer Science (Summa Cum Laude) — *Graduated 2024*

HONORS & AWARDS

- 2024 UARK Razorback Classic & Honors College Fellow
- 2024 President's Volunteer Service Award
- 2023 NSF REU Recipient
- 2022 SURF Grant Recipient
- 2020 U.S Presidential Scholar & National Merit Scholar

TECHNICAL PROFICIENCIES

- ***Languages***: Python, JS/TS, C++, SQL, Java, Bash, P4, Kotlin, Swift
- ***Frameworks & Libraries***: React/Next.js, Express, React Native, Angular, Spring Boot, PyTorch, RDKit
- ***Tools & Platforms***: Firebase, Git, Docker/Kubernetes, CI/CD pipelines, AWS

PEER-REVIEWED CONFERENCE PUBLICATIONS

- J. Norris, D. Galeano. "Benchmarking ML Models for Antibiotic Bioactivity Prediction*", Ibero-American Conference on Artificial Intelligence,* November 2026
- J. Norris, Z. He, Y. Qu, G. Chen, C. Hertzog, D. Jin. "An In-Network Approach for PMU Missing Data Recovery with Data Plane Programmability", *IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids*, October 2025 (DOI: 10.1109/smartgridcomm65349.2025.11204568)

RESEARCH EXPERIENCE

FIUNA AI/ML Drug Discovery Lab — **Asunción, Paraguay**
Research Fellow under Dr. Diego Galeano — *2025 – 2026*

- Designed and evaluated models using PyTorch, Chemprop, and Scikit-learn for predicting bioactivity
- Published paper in IBERAMIA '26 while serving in the Peace Corps

UARK CSCE Computer Networks Lab — **Fayetteville, AR**
Undergraduate Researcher under Dr. Kevin Jin — *2022 – 2024*

- Developed P4-based approaches to improve power grid security through programmable networks
- Published paper in IEEE SmartGridComm 2025; presented research poster at IEEE Greentech 2024

PROFESSIONAL EXPERIENCE

Peace Corps Paraguay — **Caazapá, Paraguay**
Community Health Volunteer — *2024 – 2026*

- Co-built and sourced a homemade community gym and 7-unit school computer lab
- Orchestrated (in Spanish) 5000lb book donation for 3 community libraries in rural Paraguay

Compound Foundry — **Fayetteville, AR**
Junior Software Engineer — *2023 –2024*

- Utilized NextJS, React, and GraphQL to rapidly prototype business ideas at local startup studio
- Built Shopify extension with Next.js that automated order printing & shipping workflows for artists

Amazon — **Seattle, WA & Los Angeles, CA**
Software Development Engineer Intern — *2021 & 2022*

- Built a no-code interface with React and Scala for product managers to modify JSON configurations without direct engineering support, reducing turnaround for internal config changes by 60%
- Implemented SharePlay into Amazon Music with Swift, achieving sub-second sync latency on up to 6 devices.

J.B Hunt Transport Services **Fayetteville, AR**
Application Development Intern *2019 –2022*

- Shipped production code using Angular, SpringBoot, and React across 3 applications (carrier management platform, internal interview feedback tool, driver mobile app), with unit tests supporting 80% code coverage

PROFESSIONAL DEVELOPMENT & LEADERSHIP

UARK Computer Science Club (ACM) **Fayetteville, AR**
President, former Vice President *2022 –2024*

- Aided faculty in organizing programming competitions and "Hour of Code" events for local schools
- Orchestrated "HogHacks", the largest hackathon in the Northwest Arkansas area, garnering $13k in funding, ~200 participants, and involvement with Major League Hacking (MLH)

McMillon Innovation Studio **Fayetteville, AR**
Technical Consultant, Senior Project Lead *2020 –2024*

- Built prototypes for business ideas from UARK's student innovation lab
- Co-led a startup that won $20k in the Arkansas Governor's Cup business plan competition
- Led projects for stakeholders of Walmart, UARK, Phigenics, Ox, and the NWA Food Bank

First-Year Engineering Program **Fayetteville, AR**
Peer Mentor *2021 –2024*

- Met weekly during the school year with 10+ freshman engineers to offer guidance, advice, and connections as they navigated their freshman years

SERVICE & VOLUNTEER WORK

UARK Volunteer Action Center **Fayetteville, AR**
Be BOLD Chair *2022 –2024*

- Led UARK sponsored middle-school mentorship program during senior year of college
- Planned and executed weekly lesson plans that briefed 15+ middle-school boys on how to code, change a tire, manage a budget, and prepare healthy meals

NWA Children's Hospital **Springdale, AR**
Volunteer *2022 –2023*

- Volunteered in the playroom of NWA Children's Hospital, entertaining children receiving treatment
- Accompanied families to pre-op rooms and provided complimentary refreshments and comfort amenities

Mission: Mentor **Boston, MA (remote)**
Volunteer Developer *2021 –2022*

- Dedicated 15 hours weekly to non-profit EdTech startup incubated at the Harvard Innovation Lab
- Co-led a team of 4 to build out and maintain the primary service with React, SpringBoot, and SQL

SELECTED PROJECTS

AI/ML for Antibiotic Discovery

- Conducted systematic comparison of 5 model architectures across 6 molecular representations on the antibiotic dataset provided by Stokes et al. (DOI: 10.1016/j.cell.2020.01.021)
- Investigated model performance when subjected to stringent data splitting methods and Bayesian hyperparameter optimization; found optimized classical baselines rivaling GNN performance

Lightweight Automatic Differentiation Engine

- Built a lightweight automatic differentiation engine in Python, implementing computational graphs & backpropagation from scratch; trained a custom neural network achieving 87.5% accuracy on MNIST dataset

Personal Portfolio & Custom Blog

- Built a responsive React website to showcase projects and host a blog chronicling personal project development and my Peace Corps service
- Developed custom graphics for Conway's Game of Life, Boids, and the three-body problem simulations

Vector Network Analyzer (VNA) App

- Built a mobile application for a UARK nanotechnology lab that interfaced via Bluetooth with a VNA used for pathogen detection in communal spaces.